Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 04)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain the meaning of each of the following terms and explain why such species are given conservation priority.

[8 marks]

EDGE species .....

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Endemic species .....

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Keystone species .....

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Flagship species .....

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0 1 . 2 Explain why the degree of population dispersal is considered when setting priorities.

[2 marks]

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0 2 Table 1 compares extensive oyster farming with intensive salmon farming.

Table 1

Feature	Extensive oyster farm	Intensive salmon farm
Food input	None - filter natural plankton	Fishmeal / plant pellets
Stocking density	Low, spaced on seabed	High, in sea cages / tanks
Breeding	Wild spat settle on cleared seabed or hatchery	Selected adults stripped of eggs and milt
Disease / pest control	Removal of predatory crabs	Antibiotics, pesticides, wrasse, removal of diseased fish
Yield per hectare	Low	Very high

0 2 . 1 Use Table 1 to explain the relative merits of extensive and intensive aquaculture.

[4 marks]

0 2 . 2 Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

Do not write outside the box

0 2 . 3 Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

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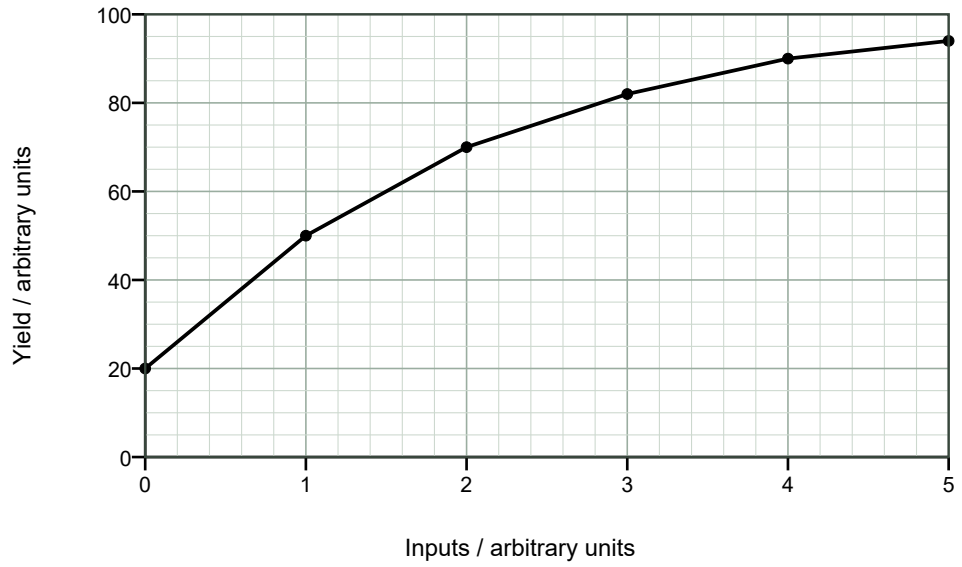
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0 3 Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1



0 3 . 1 Describe the relationship shown in Figure 1.

[2 marks]

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0 3 . 2 Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]

Answer _____ units

0 3 . 3 Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

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0 3 . 4 Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

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0 3 . 5 Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

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0 4 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

0 4 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

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0 5 . 1 Describe how each of the following abiotic factors can be managed to increase biodiversity in a nature reserve.

[6 marks]

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Light .....

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Dissolved oxygen .....

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Nutrients .....

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0 5 . 2 Explain how biotic factors can be managed to increase the survival of a species. Give **four** examples.

[4 marks]

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0 6 The Rio Bravo Conservation and Management Area in Belize is a largely undamaged rainforest with over 200 species of tree, 400 species of bird and 70 species of mammal. It is managed by a Belizean NGO to conserve wildlife, produce resources, sequester carbon and benefit the local economy.

0 6 . 1 Explain why creating the reserve could have threatened local communities, and describe how the project avoids this.

[3 marks]

0 6 . 2 Explain **four** of the rules followed when trees are selected and felled that keep the exploitation sustainable.

[4 marks]

0 6 . 3 Each felled tree is given a unique identification tag. Explain how this supports Forest Stewardship Council accreditation.

[2 marks]

Do not write outside the box

0 6 . 4 Describe **three** ways the reserve raises money without damaging the forest.

[3 marks]

0 6 . 5 Explain why selective logging of this kind causes much less ecological damage than clear felling.

[3 marks]

15

0 7 . 1 Explain why a moderate water shortage reduces crop productivity even though the plants survive.

[3 marks]

Do not write outside the box

0 7 . 2 Explain the problems caused by waterlogged soils and describe **two** ways farmers reduce soil water levels.

[4 marks]

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0 7 . 3 Explain how irrigating with groundwater can cause soil salinisation and how this harms crops.

[2 marks]

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0 7 . 4 State how a farmer could raise the pH of an acidic soil and explain why a low pH reduces fertility.

[1 mark]

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Do not write outside the box

0 8 . 1 Explain how GPS mapping, drone surveys and computer-based yield monitoring can reduce the environmental impacts of agriculture.

[3 marks]

0 8 . 2 Explain why productive agricultural technologies are less available in LEDCs.

[2 marks]

0 8 . 3 Explain how buying fairtrade food affects producers.

[1 mark]

0 8 . 4 Explain how nitrate from farmland can threaten human health.

[2 marks]

0 8 . 5 Give **two** agricultural sources of methane.

[2 marks]

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1 .....

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2 .....

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0 9 . 1 Which of the following habitats has low biodiversity but a high proportion of endemic species? Tick one box.

[1 mark]

A Tropical rainforest ☐

B Oceanic islands ☐

C Temperate broadleaf woodland ☐

D Mangroves ☐

0 9 . 2 Compare the ecological features of tropical rainforest and Antarctica that explain their very different biodiversity.

[4 marks]

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1 0 . 1 Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

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Extra space

1 0 . 2 State what is meant by a **taxon**.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

OR

11 . 2 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS